RESEARCH CONCEPT / CHICAGO METROPOLITAN AREA

Geo Infra Lens AI

Predicting Road Damage Before It Happens

A proposed early-warning approach to climate-related pavement deterioration

GeoAI • GIS • Pavement conditions • Climate data

ADNAN GHAFFAR

01 / WHAT IS OUR RESEARCH ABOUT?

Road damage is often found after it occurs.

Cracks, potholes and deteriorating surfaces can worsen following changing weather conditions.

01

FREEZE & THAW

Repeated temperature changes

02

HEAVY RAIN

Water entering pavement

03

FLOODING

Extended water exposure

04

EXTREME HEAT

Heat and traffic stress

The research challenge: identify roads at risk early enough for timely inspection.

02 / OUR RESEARCH AIM

Which roads are likely to be damaged next?

> Our aim is to predict which roads are likely to deteriorate before serious problems develop.

The intended result is a practical way for cities to find vulnerable road sections and inspect them at the right time.

03 / HOW IT MAY WORK

A winter road in Naperville

A simple example of how weather and existing pavement weakness can interact.

1

Temperatures fluctuate

Air temperature repeatedly rises above and falls below freezing.

2

Snow melts and water enters

Moisture may reach existing pavement cracks or vulnerable layers.

3

Damage may progress

Freezing, thawing and traffic can contribute to more severe distress.

4

AI estimates risk

The proposed model flags road sections for closer inspection.

04 / HOW THE AI WILL WORK

Four sources of information

The model will learn patterns from weather, roads, geography and road use.

01 **WEATHER**

Rain, snow, freezing temperatures and extreme heat.

02 **ROAD CONDITION**

Existing cracks, road age and previous repairs.

03 **GIS DATA**

Locations, elevation, drainage and environmental context.

04 **TRAFFIC**

How heavily each road section is used.

05 / FROM INFORMATION TO ACTION

Estimate risk over 30–90 days

A future risk estimate—not a claim that road damage is certain.

01

INPUT

Historical weather, road and GIS data

→

02

MODEL

Estimate risk of worsening pavement

→

03

PRIORITIZE

Identify road sections needing review

→

04

VERIFY

Inspect and compare with real outcomes

Example: identify 10 sections for review after a winter storm rather than surveying hundreds without prioritization.

06 / WHAT MAKES OUR RESEARCH DIFFERENT?

From finding damage to anticipating risk

The scientific value must be tested against existing pavement assessment methods.

TODAY: DETECT

Use cameras, reports and inspections to identify cracks and potholes already present.

PROPOSED: FORECAST

Combine weather, GIS and pavement history to estimate where damage may worsen next.

We will evaluate whether the proposed approach improves the accuracy and usefulness of inspection decisions.

07 / WHAT WE NEED TO PROVE

Three research questions

These questions define the experiments—not promises of performance.

01 **Can AI predict which roads are more likely to deteriorate after extreme weather?**

02 **Can we measure how reliable the predictions are?**

03 **Can cities inspect roads more efficiently without overlooking serious damage?**

We will compare our methods with existing models and verify findings using real observations.

08 / WHERE WE WILL START

Chicago suburbs, Illinois

Proposed study communities: Naperville, Aurora, Oswego and nearby areas.

INITIAL STUDY AREA

Naperville
Aurora
Oswego

PLANNED DATA

Historical weather information
Government roadway datasets
GIS maps and pavement inspection records

Academic and transportation engineering partners would help evaluate the methodology.

09 / WHAT WE WANT TO ACHIEVE

Better decisions, earlier

Expected benefits must be confirmed through actual field testing.

- Identify road sections at higher risk
- Provide warnings after major weather events
- Help crews decide where to inspect first
- Support better maintenance planning
- Test whether inspections and costs can be reduced

10 / LONG-TERM VISION

Change the question cities ask.

TODAY

“Which roads are damaged and need repairs?”

OUR RESEARCH ASKS

“Which roads may be damaged next—and where to inspect?”

Start in Illinois. Test the methods scientifically. Explore whether the approach can help transportation agencies across the United States.

DISCUSSION WITH RESEARCH PARTNERS

We are seeking scientific input and field validation.

Potential areas of collaboration

- Pavement deterioration mechanisms and labels
- Experimental design and model comparison
- Independent evaluation and field observations